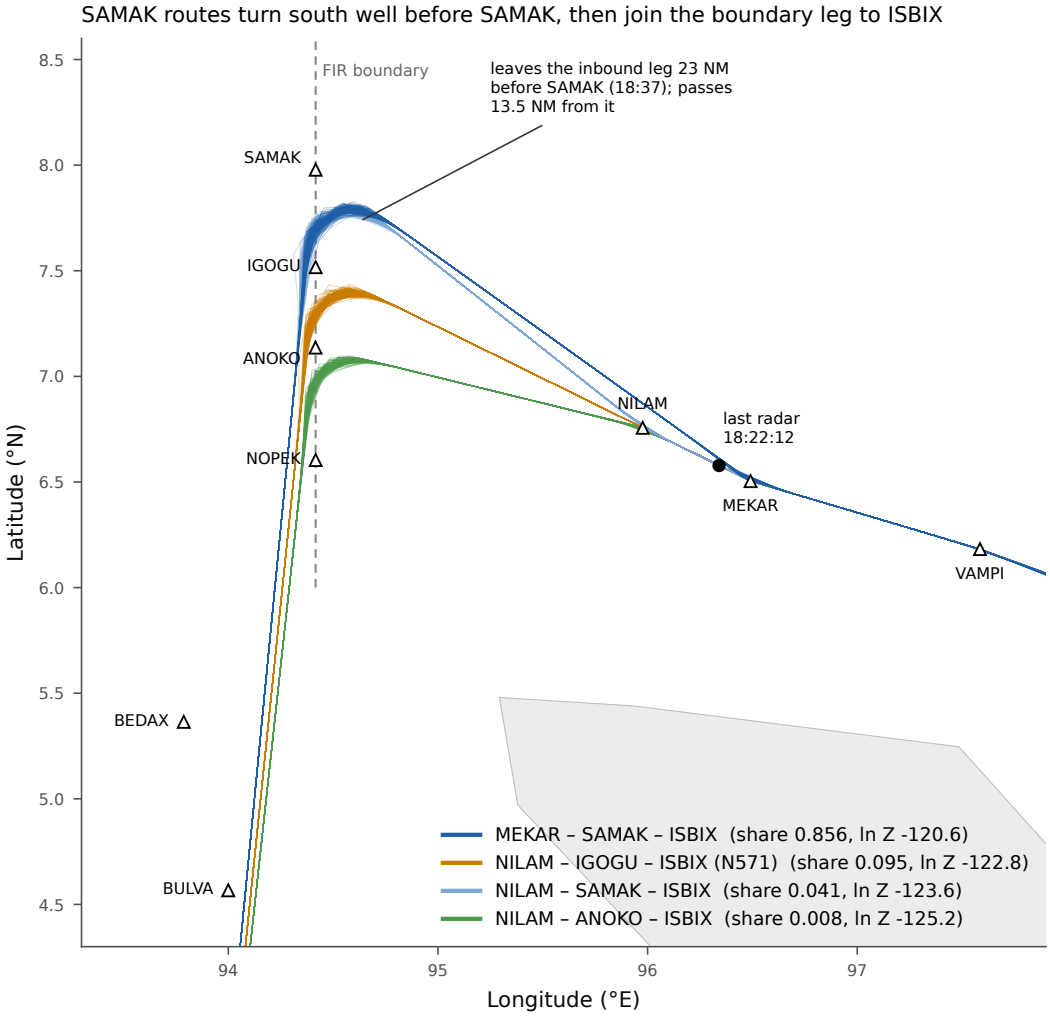



Plain note (ASD-STE100): This chart shows paths of the aircraft that follow named airway points, from the last radar point to about 19:00. Each colour is one route. In this model the aircraft starts each turn before the point (a fly-by turn), as a flight management computer does. The aircraft on the SAMAK routes does not reach SAMAK. It starts to turn south about 23 to 27 NM before SAMAK, at about 18:37, and passes about 13 NM from it. Then it flies south near the FIR boundary to ISBIX. The two routes that end at BULVA are not shown: the data give them almost no probability. The chart shows only the airway points in our table. The published chart has more points in this area that we did not sample.

Technical note: paths re-flown locally for display (binary 409192d, fly_by + track_legs + turn_backs, run H stack: davey2016-inmarsat, no-exhaustion-prior, s1-s6 fuel, reference-snapshots, radar-full, s7/s8 PROVISIONAL, inop-flow-fix), 1 seed x 84k particles per route, 400 posterior route samples per route at 100 s spacing (120 drawn), observations after m1941 excluded and fuel power required to m1941 only, to allow 100 s spacing under the 64-vertex limit; geometry only, not used for evidence. Shares and ln Z in the legend are run H (8 seeds x 3M, route identifier, equal prior 1/6 over routes 10, 11, 3, 2, 19, 43). Turn start: first point after 18:31:49 more than 0.5 NM off the inbound great-circle leg (SAMAK routes 23.3-23.7 NM before the fix at 18:37:06-18:37:11; back-projected along a 13 NM arc, 26-27 NM at 18:36:40-18:36:45); closest approach from paths subdivided 40x between vertices; theoretical fly-by anticipation R tan(dpsi/2) with R = 13 NM (15 deg bank, ~490 kt): SAMAK 120 deg 22.6 NM, IGOGU 110 deg 18.6 NM, ANOKO 97 deg 14.7 NM; closest approach R(sec(dpsi/2) - 1): 13.0 / 9.7 / 6.6 NM against 13.5 / 9.5 / 6.8 NM measured. Waypoints: engine/data/waypoints.csv (third-party navdata, current cycle, epoch not verified for 2014). FIR boundary drawn along 094 deg 25 min E. Land: Natural Earth 1:110m.